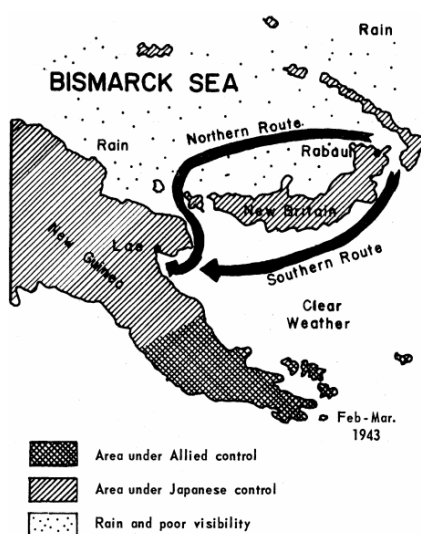


Lesson 8.1: Introduction to games

Introduction to game theory

Victor Jauregui

Battle of the Bismarck Sea



Players

Decision-makers involved:

- ▶ Allies' General
- ▶ Convoy Captain

Outcomes/payoffs

Number of days convoy attacked
(0–3)

Question

How will this situation (game) play out?

Game models

	n	s
N	2	2
S	1	3

row player Allies' General

column player Convoy Captain

Preferences

What are best and worst outcomes for each player?

row player *maximise* number of days convoy attacked

column player *minimise* number of days convoy attacked

or *maximise* number of days convoy not attacked

	n	s
N	2-1	2-1
S	1-2	3-0

Solutions

Question

How will this situation (game) play out?

	n	s
N	2	2
S	1	3

	n
N	2
S	1

	n
N	2

Solution

Rational *play* is (N, n). Value is 2.